AMAZON BRAZIL ANALYSIS REPORT

Amazon, a global leader in e-commerce, has achieved significant success in markets like the U.S., Europe, and Asia. In Brazil, Amazon connects small and medium businesses with millions of customers, becoming a key player. Given the similarities between Brazil and India—such as large populations and diverse consumer bases—there's an opportunity to replicate success in India.

ANALYSIS I

Question 1: Round the average payment values to integer (no decimal) for each payment type and display the results sorted in ascending order.

Context: Amazon India wants to standardize financial reporting by removing decimal variations in average payment values. The objective is to compute the average payment amount for each payment type, round it to the nearest integer, and present the results in ascending order for simplified comparison.

SQL Query (PostgreSQL) snippets:

```
--Q1 Avg payments by payment_type
select payment_type,ROUND(AVG(payment_value)) AS rounded_avg_payment
from amazon_brazil.payments
GROUP BY payment_type ORDER BY rounded_avg_payment Asc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	payment_type character varying	rounded_avg_payment numeric
1	not_defined	0
2	voucher	66
3	debit_card	143
4	boleto	145
5	credit_card	163

Total rows: 5 | Query complete 00:00:00.108 | CRLF | Ln 21, Col 1

Analysis :-

From the output:

- Not_defined shows very low average (0) — possibly incomplete or system-generated entries.
- voucher has the lowest valid average payment.
- credit card has the highest average payment value.

- This shows that customers using credit cards tend to make higher-value transactions compared to other payment methods

Recommendation:

- Identify not_defined payment type to ensure data accuracy and remove potential anomalies.
- Promote credit card partnerships and cashback programs, as this payment method generates higher average revenue.
- Consider targeted marketing strategies for voucher and debit card users to increase their average transaction value.

Question 2: Percentage of Orders by Payment Type

Context: Amazon wants to analyze how orders are distributed across different payment methods to refine its payment strategy.

SQL Query (PostgreSQL) snippets:

```
--Q2 Total orders percentage and payment_type
SELECT payment_type,ROUND(COUNT(order_id)*100.0/(Select COUNT(*) from amazon_brazil.payments),1)||'%' A
GROUP BY payment_type ORDER BY total_percentage_orders Desc;
--From this analysis we can see that te credit card percent is dominating so we can take this insights
--credit card followed by boleto
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	payment_type character varying	total_percentage_orders text
1	credit_card	73.9%
2	voucher	5.6%
3	boleto	19.0%
4	debit_card	1.5%
5	not_defined	0.0%

Total rows: 5 | Query complete 00:00:00.113 | CRLF | Ln 35, Col 1

Analysis:

- Credit Card (~73.9%) dominates total orders.
- Boleto (~19.0%) is the second most used.
- Voucher and debit card have significantly smaller shares.
- not_defined shows negligible usage.
- This clearly shows strong customer preference for credit card payments

Recommendation:

This query calculates customer-level average order value and ranks them using a window function. Top-ranked customers can be targeted for loyalty rewards and exclusive offers.

Question 3: Find all products priced between 100–500 BRL **that contain** “Smart” **in their name.**

Context: Amazon India wants to create targeted promotions for mid-range smart products.

SQL Query (PostgreSQL) snippets:

```
--Q3 product_id & Price
SELECT p.product_id, oi.price FROM amazon_brazil.products p
JOIN amazon_brazil.order_items oi on p.product_id = oi.product_id
WHERE oi.price BETWEEN 100 AND 500 AND p.product_category_name LIKE '%Smart%'
ORDER BY oi.price Desc;
```

Data Output Messages Notifications

product_id character varying	price numeric

Total rows: 0 Query complete 00:00:00.109 CRLF Ln 39, Col 1

Analysis:

No any result found , there is some language no any language starting with ‘Smart’

Question 4: Top 3 Highest Sales Months Analysis

Context: Amazon India wants to identify seasonal sales trends by analyzing monthly performance and focusing on the months generating the highest total sales.

SQL Query (PostgreSQL) snippets:

```sql
--Q4 total_sales & month
SELECT EXTRACT(MONTH FROM o.order_purchase_timestamp)AS month,ROUND(SUM(oi.price)) AS total_sales
FROM amazon_brazil.orders o JOIN amazon_brazil.order_items oi on o.order_id =oi.order_id
GROUP BY month ORDER BY total_sales DESC Limit 3;
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	month numeric	total_sales numeric
1	1	50457
2	3	48905
3	5	44829

Business Insights:

- Month 1 generated the highest revenue, indicating strong beginning-of-year purchasing behavior.
- Months 1 and 3 both appear in the top 3, showing that Q1 is a high-performing quarter.
- Sales slightly decreased from Month 1 to Month 5, suggesting demand softening as the year progresses.

Recommendation:

- **Increase Marketing Budget in High-Performing Months**
  Allocate more promotions and ads in Months 1 and 3 to maximize revenue during peak demand.

- **Replicate Q1 Strategy in Lower Months**
  Analyze campaigns, discounts, and product categories from Month 1 and apply similar strategy in weaker months.

- **Launch Pre-Peak Campaigns**
  Start promotional activities 2–3 weeks before Month 1 to capture early buying intent and increase overall peak sales.

Question 5: High Price Variation Category Analysis

Context: Amazon India wants to identify product categories that have significant price differences. Large price variation within a category may indicate diverse product positioning (budget to premium), competitive intensity, or upselling opportunities.

SQL Query (PostgreSQL) snippets:

```sql
--Q5 product-category-name with price differnce(Max-min)

SELECT p.product_category_name, MAX(oi.price)-MIN(oi.price) AS price_difference FROM
amazon_brazil.products p JOIN amazon_brazil.order_items oi
on p.product_id = oi.product_id GROUP BY p.product_category_name
HAVING MAX(oi.price)-MIN(oi.price)>500
ORDER BY price_difference DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 57 Page No: 1 of 1

	product_category_name character varying	price_difference numeric
2	pcs	6694.5
3	artes	6495.5
4	eletroportateis	4792.5
5	instrumentos_musicais	4394.97
6	consoles_games	4094.81
7	esporte_lazer	4054.5
8	relogios_presentes	3990.91

Total rows: 57 Query complete 00:00:00.213 CRLF

Business Insights:

- Categories like *pcs* and *artes* show extremely high price gaps, indicating a mix of low-end and premium products within the same category.
- Categories such as *electronics and gaming* (eletroportateis, consoles_games) naturally support premium pricing, suggesting strong upsell potential.
- Large price differences may indicate multiple sellers, varied brands, and product differentiation within the same category.

Recommendation:

- **Segment Categories by Price Tier**
  Create clear sub-segments (Budget / Mid-range / Premium) for high-variation categories to improve targeting and recommendation algorithms.

- **Upsell Strategy in High-Variation Categories**
  Promote premium versions of products in categories like PCs and gaming through bundles, EMI offers, or “recommended upgrade” tags.

- **Targeted Promotions for Lower-End Products**
  Use discounts and flash sales for lower-priced items to increase volume, while maintaining margin on premium products.

Question 6: Payment Consistency Analysis

Context: Amazon India aims to improve customer experience by identifying payment methods that show consistent transaction behavior. Lower variation in transaction amounts indicates predictable spending patterns and stable usage.

SQL Query (PostgreSQL) snippets:

```sql
--Q6 payment-type with payment value std.deviation
SELECT payment_type, ROUND(STDDEV(payment_value),2) AS Std_deviation
FROM amazon_brazil.payments GROUP BY payment_type
ORDER BY Std_deviation ASC;
```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	payment_type character varying	std_deviation numeric
1	not_defined	0.00
2	voucher	115.52
3	boleto	213.58
4	credit_card	222.12
5	debit_card	245.79

Total rows: 5 Query complete 00:00:00.124 CRLF Ln 59, Col 41

Business Insights:

- Voucher shows the lowest standard deviation (115.52), meaning customers using vouchers tend to make similar-value purchases.
- Credit and debit cards have higher usage, indicating they are used across both small and large transactions.
- Debit card transactions fluctuate the most, suggesting diverse usage behavior and less predictable spending patterns.
- 'Not Defined' Requires Data Cleaning of either missing or single-value transactions — this category should be reviewed for data accuracy.

Recommendation:

- **Use Vouchers for Targeted Campaigns**
  Since voucher payments are consistent, Amazon can design fixed-value discount campaigns for voucher users.

- **Segment Card Users by Spending Behavior**
  Credit and debit card users should be segmented into low, mid, and high spenders for personalized offers.

- **Review and Clean 'Not Defined' Category**
  Investigate and correct missing or unclear payment type records to improve analytical accuracy.

- **Leverage Credit Card for High-Value Upselling**
  Because credit cards show high variability, they are ideal for EMI offers, premium product promotions, and bundle deals.

Question 7: Retrieve products where the product category name is missing (NULL) or contains only a single character.

Context: Amazon India aims to improve catalog accuracy and customer experience by identifying products with incomplete or invalid category names. Missing or incorrect category labels can impact search results, recommendations, and reporting accuracy.

SQL Query (PostgreSQL) snippets:

```sql
--Q7 product_id & product-category-name (missing name & single cahracter)
SELECT product_id, product_category_name FROM amazon_brazil.products
WHERE product_category_name IS NULL OR LENGTH(TRIM(product_category_name))=1;
```

Data Output Messages Notifications

Showing rows: 1 to 614 Page No: 1 of 1

	product_id [PK] character varying	product_category_name character varying
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdcbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]

Total rows: 614 Query complete 00:00:00.132 CRLF Ln 63, Col 1

Business Insights:

- 614 products have missing or invalid category names, which can directly affect search accuracy, filtering, and recommendation engines.
- Products without proper categories may not appear in relevant browsing sections, reducing visibility and potential sales.
- Category-level sales analysis, demand forecasting, and inventory planning may be inaccurate due to unclassified products.Debit card transactions fluctuate the most, suggesting diverse usage behavior and less predictable spending patterns.
- Missing category tags reduce search engine optimization and internal search relevance.

Recommendation:

- Immediate Data Cleansing Initiative
  Assign missing categories using product metadata, historical patterns, or AI-based classification models.

- Implement Validation Rules
  Add database constraints to prevent NULL or single-character category names during product upload.

- Automated Category Suggestion System
  Use machine learning or rule-based mapping to auto-assign categories when sellers list products.

- Seller Compliance Monitoring
  Notify sellers whose products lack proper categorization and restrict publishing until corrected.

ANALYSIS II

Question 1: Segment order values into three categories: Low Value: < 200 BRL ,Medium Value: 200 – 1000 BRL,High Value: > 1000 BRL
Calculate the count of each payment type within these segments and display results in descending order of count.

Context: Amazon India wants to analyze how payment preferences vary across different order value segments. Understanding this behavior helps in designing targeted payment offers, cashback strategies, and checkout optimizations.

SQL Query (PostgreSQL) snippets:

```sql
--Q1 order_value_segment, payment_type, count
SELECT payment_type, CASE When payment_value <200 THEN 'Low'
WHEN payment_value BETWEEN 200 AND 1000 THEN 'Medium'
ELSE 'High'
END AS order_value_segment,COUNT(*) AS count
FROM amazon_brazil.payments
GROUP BY order_value_segment,payment_type
ORDER BY count DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 13 Page No: 1 of 1

	payment_type character varying	order_value_segment text	count bigint
1	credit_card	Low	60548
2	boleto	Low	16444
3	credit_card	Medium	15303
4	voucher	Low	5476
5	boleto	Medium	3162
6	debit_card	Low	1287
7	credit_card	High	944

Total rows: 13 Query complete 00:00:00.118 CRLF Ln 74, Col 1

Business Insights:

- Credit card is the most preferred payment method in Low and Medium order value segments, and likely continues in High as well.This shows customers prefer flexibility, EMI options, and reward-based payments.
- Boleto has significant usage in the Low segment, but drops in Medium and High ranges.This suggests price-sensitive customers prefer boleto for smaller purchases.
- Higher order values are more likely completed via credit card, indicating trust in online digital transactions for expensive purchases.

Recommendation:

- Promote EMI & Cashback on Credit Cards -Since credit card dominates, Amazon should Offer EMI options on Medium & High segments .Provide cashback/loyalty

rewards .Partner with banks for exclusive deal

- Optimize Checkout Experience by Segment-Highlight credit card offers in High-value carts.Show boleto as default option in Low-value carts (if behaviorally relevant).Reduce payment friction for dominant methods

- Target Boleto Users with Upsell Strategies- For customers using boleto in Low segment:Provide small discounts for upgrading to credit card.Introduce first-time card usage incentives.Offer bundle deals to increase cart value

Question 2: Calculate the minimum, maximum, and average price for each product category and list them in descending order of average price.

Context: Amazon India wants to understand pricing distribution across product categories. Analyzing minimum, maximum, and average prices helps identify premium categories, competitive segments, and pricing strategy opportunities.

SQL Query (PostgreSQL) snippets:

```
--Q2 product_category_name , min_price,max_price,avg_price
SELECT p.product_category_name, ROUND(MIN(oi.price),1) AS min_price,ROUND(MAX(oi.price)) AS max_price,
ROUND(AVG(oi.price),2) AS avg_price FROM amazon_brazil.products p JOIN amazon_brazil.order_items oi on
GROUP BY product_category_name ORDER BY avg_price DESC;
```

Data Output Messages Notifications

Showing rows: 1 to 79 Page No: 1 of 1

	product_category_name character varying	min_price numeric	max_price numeric	avg_price numeric
1	pcs	34.5	6729	1098.34
2	portateis_casa_forno_e_cafe	10.2	2899	624.29
3	eletrodomesticos_2	13.9	2350	476.12
4	agro_industria_e_comercio	13.0	2990	341.66
5	instrumentos_musicais	4.9	4400	281.62
6	eletroportateis	6.5	4799	280.78
7	portateis_cozinha_e_preparadores_de_aliment...	17.4	1099	264.57

Total rows: 79 Query complete 00:00:00.241 CRLF Ln 79, Col 1

Business Insights:

The category pcs has the highest average price (~1098) and the highest maximum price (~6729). Other categories such as portateis_casa_forno_e_cafe and eletrodomesticos_2 also show relatively high average prices. These categories represent **high-value product segments**, meaning fewer sales can still generate significant revenue.

Several categories show a **large gap between minimum and maximum prices**.This indicates that these categories contain **both low-cost accessories and expensive main products**, leading to a wide pricing range.

Example:

- pcs → Min: **34.5**, Max: **6729**
- instrumentos musicais → Min: **4.9**, Max: **4400**
- Categories such as: instrumentos musicais,eletroportáteis ,cozinha/preparadores have moderate average prices (~250–300).Products in this price range are generally **more affordable for customers**, suggesting they may contribute to **higher order volumes**.
- Some categories have **very low minimum prices**, for example **4.9** in musical instruments. These could represent **accessories, add-ons, or promotional products** used to attract customers and increase cart value.
- Many top categories in the output belong to **electronics, appliances, or household equipment**.
  This suggests that the business may be **strongly positioned in technology and home-related product segments**.

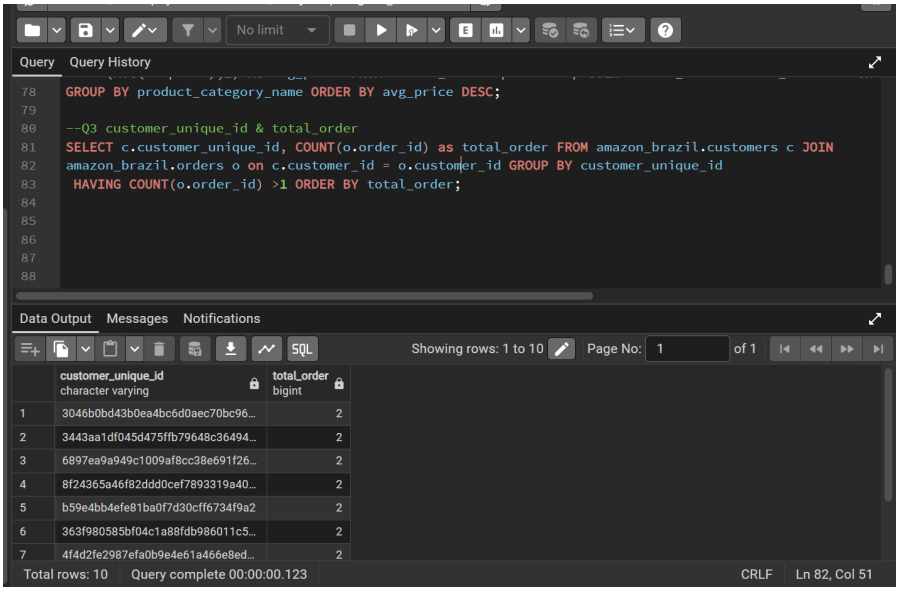
Recommendation:

- Since categories like pcs and home appliances have high average prices, promoting them through targeted marketing campaigns and featured listings could significantly increase overall revenue.Due to the large price variation within categories, products should be segmented into:
- Budget
- Mid-range
- Premium
- This allows the business to **target different customer segments more effectively**.
- Products in the **mid-price range** are ideal for **bundle offers** (e.g., appliances with accessories or instruments with related equipment).This can increase the **average order value per customer**.
- For high-priced items such as electronics or PCs, offering **installment payment options or financing** can improve conversion rates and encourage customers to purchase premium products.

Question 3: Find customers who have placed more than one order and calculate the total number of orders they have placed.

Context:Amazon India wants to identify customers who have made multiple purchases over time. Recognizing repeat customers helps in loyalty program design, retention strategies, and personalized marketing campaigns.

SQL Query (PostgreSQL) snippets:



```
GROUP BY product_category_name ORDER BY avg_price DESC;

--Q3 customer_unique_id & total_order
SELECT c.customer_unique_id, COUNT(o.order_id) as total_order FROM amazon_brazil.customers c JOIN
amazon_brazil.orders o on c.customer_id = o.customer_id GROUP BY customer_unique_id
 HAVING COUNT(o.order_id) >1 ORDER BY total_order;
```

	customer_unique_id character varying	total_order bigint
1	3046b0bd43b0ea4bc6d0aec70bc96...	2
2	3443aa1df045d475ffb79648c36494...	2
3	6897ea9a949c1009af8cc38e691f26...	2
4	8f24365a46f82ddd0cef7893319a40...	2
5	b59e4bb4efe81ba0f7d30cff6734f9a2	2
6	363f980585bf04c1a88fdb986011c5...	2
7	4f4d2fe2987efa0b9e4e61a466e8ed...	2

Total rows: 10 Query complete 00:00:00.123 CRLF Ln 82, Col 51

Business Insights:

- Repeat customers usually contribute significantly to revenue because they already trust the platform and are more likely to purchase again.
- While repeat purchases exist, the relatively low order counts (mostly around 2) suggest that long-term customer retention might still be limited.
- Encouraging more repeat purchases from these customers can significantly increase **overall revenue without acquiring new customers**.
- Different customer segments require **different marketing strategies**.
  One time buyer , repeated customers, high frequency buyers

Recommendation:

- To encourage repeat purchases, the business can introduce:
- Loyalty points
- Rewards programs
- Exclusive discounts for returning customers
- This helps increase customer retention.

Target repeat customers with:

- Personalized product recommendations
- Email promotions
- Special offers

This increases the chances of additional purchases

Customers who have already purchased once should be targeted with:

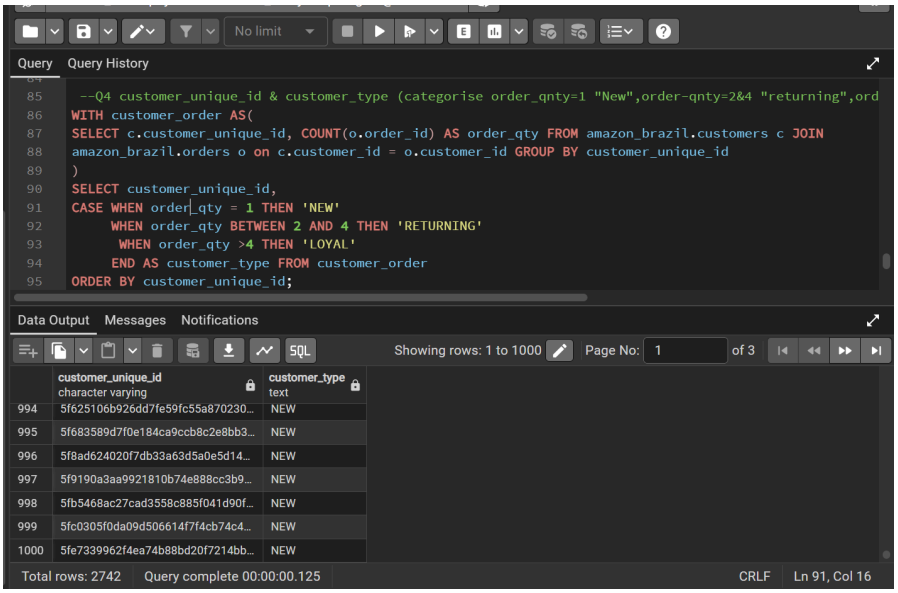
- Reminder emails
- Product suggestions
- Abandoned cart campaigns

This can convert them into frequent buyers.

Question 4: Create a temporary table to define customer segments based on total order count, then join it with the customers table to assign and display customer types.

Context:Amazon India wants to segment customers based on their purchase frequency to design personalized marketing strategies, loyalty programs, and retention campaigns.

SQL Query (PostgreSQL) snippets:



Business Insights:

Output shows many customers categorized as NEW, meaning they only purchased once.Customer acquisition is working.But retention may be weak.

If most customers remain NEW, it suggests:

- Customers are not converting into repeat buyers.
  Possible issues: shipping cost,delivery experience,product satisfaction,lack of remarketing.
- Customers with **2–4 orders** are the **most valuable opportunity segment**.Because they:already trust the brand.They have proven purchase intent
  are easier to convert into loyal customers.This segment should be **nurtured aggressively**

Recommendation:

The biggest growth opportunity is second purchase conversion.

Actions:

- Post-purchase email campaigns
- First repeat purchase discounts
- Personalized product recommendations
- Retargeting ads

Running campaigns for loyal customers should be **rewarded and protected**.

Actions:

- Loyalty program
- VIP discounts
- early access to sales
- referral rewards

Question 5: Create a temporary table to define customer segments based on total order count, then join it with the customers table to assign and display customer types.

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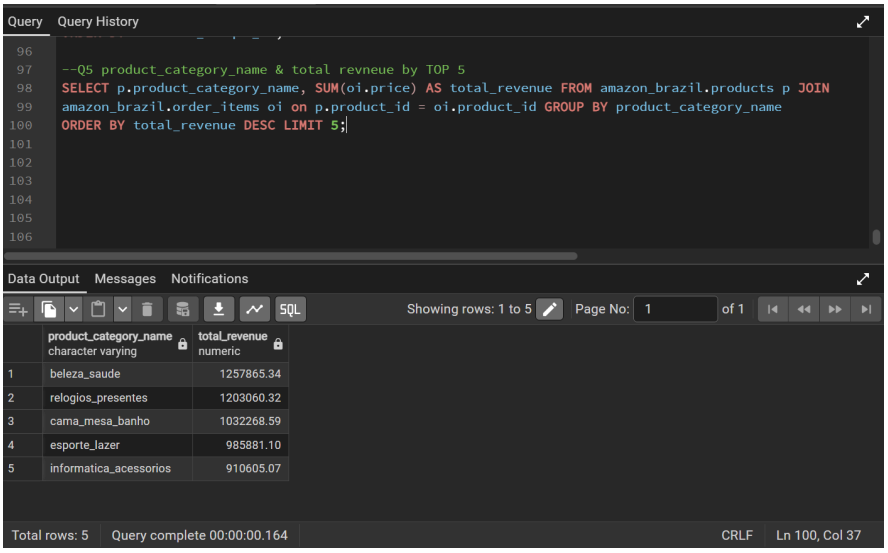
SQL Query (PostgreSQL) snippets:



Business Insights:

- beleza_saude → 1,257,865.This category generates the highest revenue among all categories.High consumer demand as there is strong repeat purchase potential. They have strong brand loyalty.
- relogios_presentes(gifts) → 1,203,060.This category is very close to the top.

- **cama_mesa_banho(bed, table, bath) → 1,032,268** This indicates strong demand for home & living products

Recommendation:

Since these categories generate the most revenue, marketing should prioritize them.Examples:

- targeted ads
- influencer promotions
- bundle offers

For **watches & gifts**, revenue can increase through seasonal campaigns.Examples:

- Valentine's Day
- Christmas
- Black Friday
- Mother's Day

ANALYSIS III

Question 1: Seasonal Sales Comparison Analysis.

Context:Amazon India's marketing team wants to compare total sales across different seasons to understand seasonal demand patterns and optimize campaign planning.

SQL Query (PostgreSQL) snippets:

```sql
--Q1 --total sales for each season (Spring, Summer, Autumn, Winter) based on order purchase dates
SELECT season, SUM(total_sales) AS total_sales
FROM(
SELECT CASE
WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN(3,4,5) THEN 'SPRING'
WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN(6,7,8) THEN 'SUMMER'
WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN(9,10,11) THEN 'AUTUMN'
ELSE 'WINTER'
END AS season,
oi.price AS total_sales FROM amazon_brazil.orders o JOIN amazon_brazil.order_items oi
on o.order_id = oi.order_id ) AS seasonal_data
GROUP BY season ORDER BY total_sales DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	season text	total_sales numeric
1	SPRING	136210.94
2	WINTER	103863.19
3	SUMMER	90744.20
4	AUTUMN	41634.78

Total rows: 4 Query complete 00:00:00.099 CRLF Ln 111, Col 15

Business Insights:

Spring is the peak revenue season for the business.Possible reasons

- Seasonal promotions
- New product launches
- Increased consumer spending
- Post-holiday shopping

  This indicates strong demand in this period.

Winter sales (~103K) are the second highest.Possible reasons:

- Holiday shopping
- Gift purchases
- Seasonal promotions

  This suggests winter campaigns likely drive strong revenue.

Recommendation:

Since spring is the top-performing season, focus on:

- aggressive marketing campaigns
- new product launches
- promotional discounts

These seasons drive the **most sales**, so stock management is critical.Recommendation:

- forecast demand
- increase inventory levels
- prioritize high-demand products

Stockouts during peak seasons could lead to significant revenue loss.

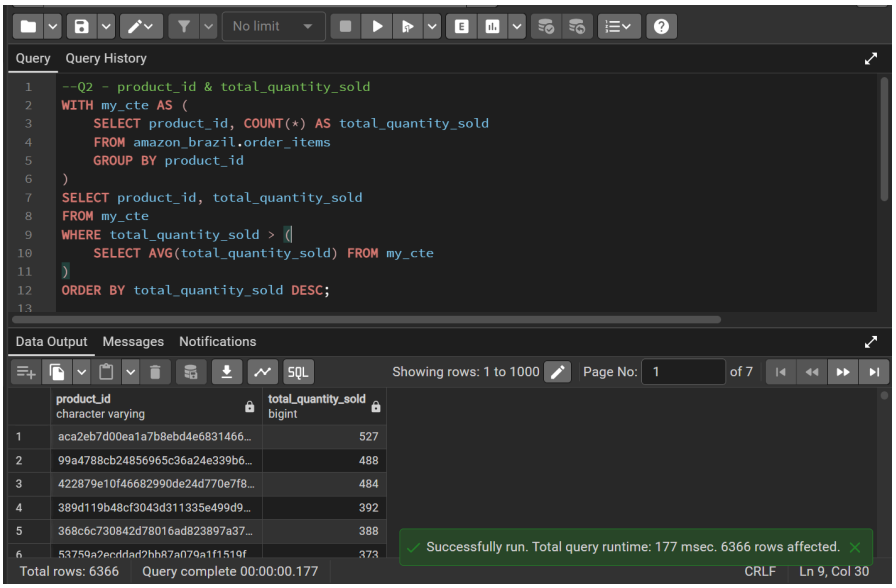
Autumn is the **lowest-performing season**.Strategies to improve sales:

- flash sales
- seasonal discounts
- clearance campaigns

Question 2: Use a subquery to calculate the overall average quantity sold across all products, and filter products whose total quantity sold is greater than the average.

Context:The inventory team wants to identify high-performing products that sell above the overall average quantity. This helps in inventory planning, stock prioritization, and demand forecasting.

SQL Query (PostgreSQL) snippets:



Business Insights:

Demand is highly concentrated in certain products

A few products are selling significantly more than others (500+ units vs average). This suggests:

- Strong customer demand
- Possibly popular categories or brands
- Potential best-seller products
  These products likely drive a large share of revenue.

High-selling products may indicate:

- strong product-market fit
- successful marketing campaigns
- favorable pricing

  These signals can guide future campaigns.

Recommendation

Prioritize inventory for best sellers

Ensure sufficient stock levels for top-performing products.Actions:

- Increase safety stock
- Improve supplier replenishment frequency
- Monitor weekly sales trends

Feature best sellers on the website

Promote high-demand products.Examples:

- Best Sellers
- "Most Popular Products"
- Homepage highlights

This can increase conversion rates.

Use top products in marketing campaigns:

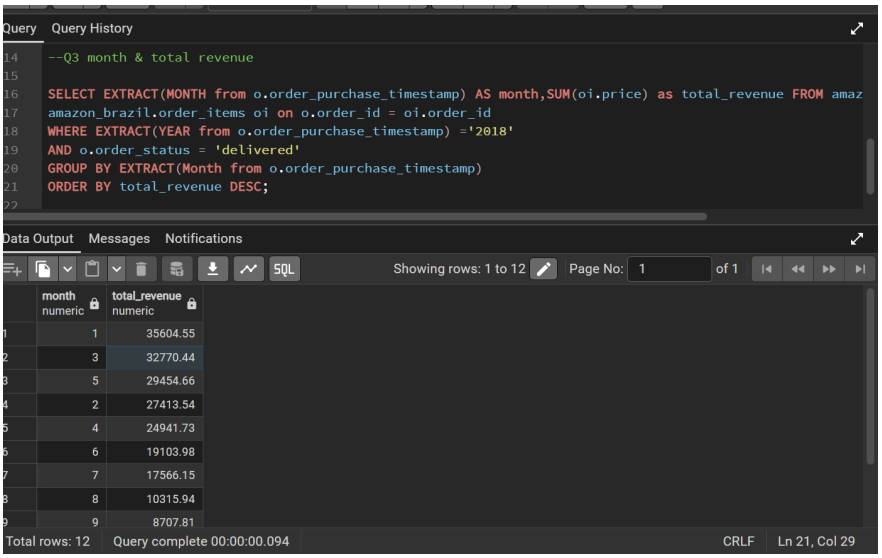
- Email promotions
- Ads
- Seasonal campaigns

These products already have proven demand.

Question 3: Calculate the total revenue generated each month in 2018.Identify months with highest and lowest revenue.Export the data to Excel.Create a line graph to visually represent monthly revenue trends

Context:The finance team at Amazon India wants to analyze monthly revenue trends for the year 2018 to identify peak and low sales periods. This helps in financial forecasting, budgeting, and seasonal campaign planning.

SQL Query (PostgreSQL) snippets:



Business Insights:

January generated the highest revenue

January produced 35.6K, the highest revenue month.

Possible reasons:

- Post-holiday shopping
- New Year promotions
- Customers spending gift cards or bonuses

Revenue stays high during Q1 and early Q2.

- Jan → Mar → May remain strong
- After June revenue drops significantly

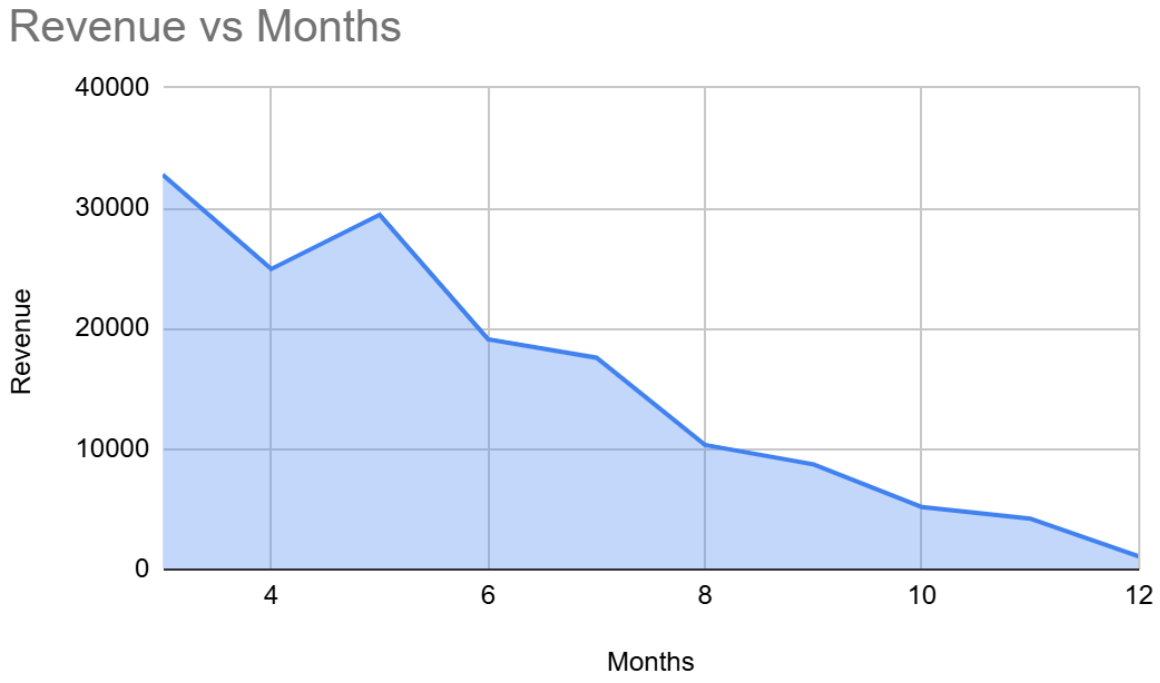
This indicates seasonal demand early in the year.

From June onwards revenue decreases.Example:

- May: **29.4K**
- June: **19.1K**
- August: **10.3K**
- September: **8.7K**

This may indicate:

- weaker seasonal demand
- reduced promotions



Recommendation

Invest more marketing in high-revenue months.Since Jan–May performs best, businesses should:

- increase advertising
- launch promotions
- run email campaigns

maximize revenue during peak demand.

Revenue drops sharply mid-year.Strategies:

- flash sales
- limited discounts
- bundle deals
- loyalty programs
  boost sales during slow periods.

Question 4: Use a CTE (Common Table Expression) to calculate total orders per customer. Classify customers into the defined segments.Count the number of customers in each segment.Export the result to Excel and create a chart showing the proportion of each segment.

Context:Amazon India is designing a loyalty program and wants to segment customers based on purchase frequency. This segmentation will help in offering personalized rewards and improving retention strategies.

SQL Query (PostgreSQL) snippets:

Query Query History

```sql
--Q4 - customer_type & count of order

WITH my_cte AS(
SELECT c.customer_unique_id,COUNT(o.order_id) as total_order FROM amazon_brazil.customers c JOIN
amazon_brazil.orders o on c.customer_id =o.customer_id GROUP BY c.customer_unique_id
),
customer_segments AS(
SELECT customer_unique_id ,
CASE WHEN total_order BETWEEN 1 AND 2 THEN 'Occassional'
WHEN total_order BETWEEN 3 AND 5 THEN 'Regular'
WHEN total_order >5 THEN 'Loyal' end as customer_type
FROM my_cte
)
SELECT customer_type,
COUNT(*) AS count FROM customer_segments
GROUP BY customer_type
ORDER BY count;
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	customer_type text	count bigint
1	Regular	3
2	Occassional	2739

Total rows: 2 Query complete 00:00:00.220 CRLF Ln 36, Col 22

Business Insights:

About 2739 customers are occasional buyers, meaning they purchased only 1–2 times.This indicates:

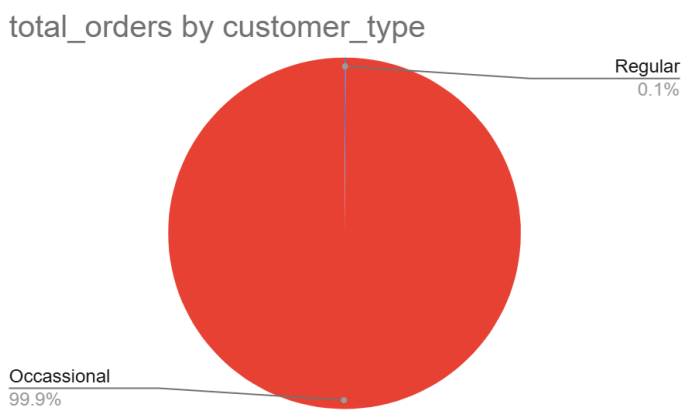
- Low repeat purchase rate
- Customers may not be strongly attached to the platform
- Acquisition may be stronger than retention

Only 3 customers are regular buyers (3–5 orders).This suggests:

- weak customer loyalty
- limited repeat purchase behavior
- customers may be price-sensitive or shopping elsewhere

There are 0 loyal customers.Possible reasons:

- dataset timeframe is short
- product categories are non-frequent purchases
- lack of loyalty programs or incentives



Recommendation

Introduce a loyalty program.Encourage repeat purchases with:

- reward points
- cashback
- exclusive discounts
- membership benefits

Use email and remarketing campaigns.Target **occasional customers** with:

- personalized product recommendations
- reminder emails
- abandoned cart campaigns

convert **occasional** → **regular buyers**.

Offer repeat purchase incentives.Examples:

- second purchase discount
- bundle deals
- free shipping on next order

These reduce friction for repeat purchases.

Improve customer experience**.**Retention improves when:

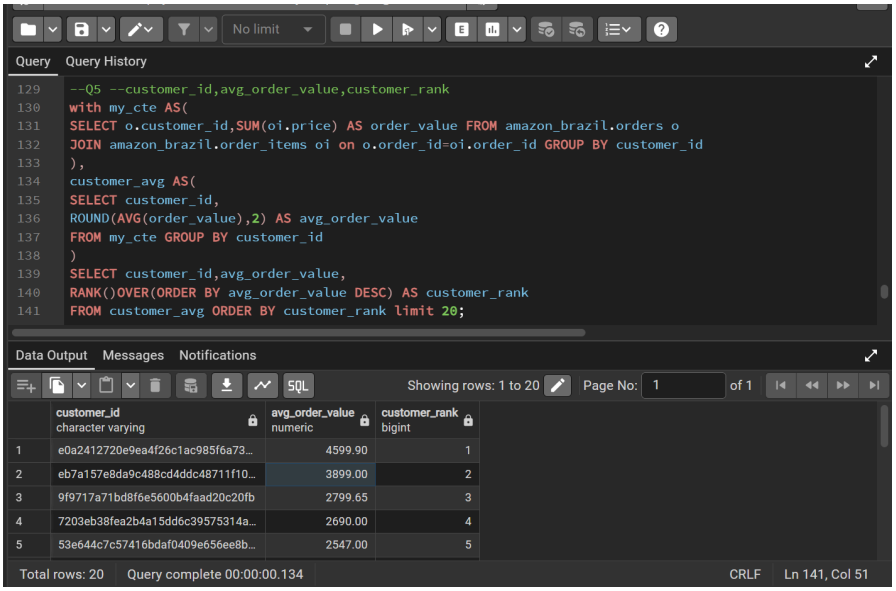
- delivery is fast
- product quality is consistent
- returns are easy

Customer experience strongly affects repeat orders.

Question 5: Calculate the average order value (AOV) for each customer.Rank customers based on their AOV in descending order.Identify the Top 20 customers with the highest average order value.

Context:Amazon India wants to identify high-value customers for an exclusive rewards program. By ranking customers based on their average order value, the company can target premium customers with personalized benefits and retention offers.

SQL Query (PostgreSQL) snippets:



```sql
--Q5 --customer_id,avg_order_value,customer_rank
with my_cte AS(
SELECT o.customer_id,SUM(oi.price) AS order_value FROM amazon_brazil.orders o
JOIN amazon_brazil.order_items oi on o.order_id=oi.order_id GROUP BY customer_id
),
customer_avg AS(
SELECT customer_id,
ROUND(AVG(order_value),2) AS avg_order_value
FROM my_cte GROUP BY customer_id
)
SELECT customer_id,avg_order_value,
RANK()OVER(ORDER BY avg_order_value DESC) AS customer_rank
FROM customer_avg ORDER BY customer_rank limit 20;
```

Showing rows: 1 to 20 Page No: 1 of 1

	customer_id character varying	avg_order_value numeric	customer_rank bigint
1	e0a2412720e9ea4f26c1ac985f6a73...	4599.90	1
2	eb7a157e8da9c488cd4ddc48711f10...	3899.00	2
3	9f9717a71bd8f6e5600b4faad20c20fb	2799.65	3
4	7203eb38fea2b4a15dd6c39575314a...	2690.00	4
5	53e644c7c57416bdaf0409e656ee8b...	2547.00	5

Total rows: 20 Query complete 00:00:00.134 CRLF Ln 141, Col 51

Business Insights:

A small group of customers generates very high order values.Your results show customers with avg order value ≈ 4599, 3899, 2799…

This indicates:

- These customers consistently place **high-value orders**
- They represent **premium/high-value customers**
- They likely drive a **disproportionate share of revenue**

Recommendation

Create a VIP Customer Program.Top ranked customers should receive:

- Exclusive discounts
- Early product access
- Loyalty rewards
- Dedicated support

Retain high revenue customers

Customers near the top but not in the top ranks could be encouraged to spend more.

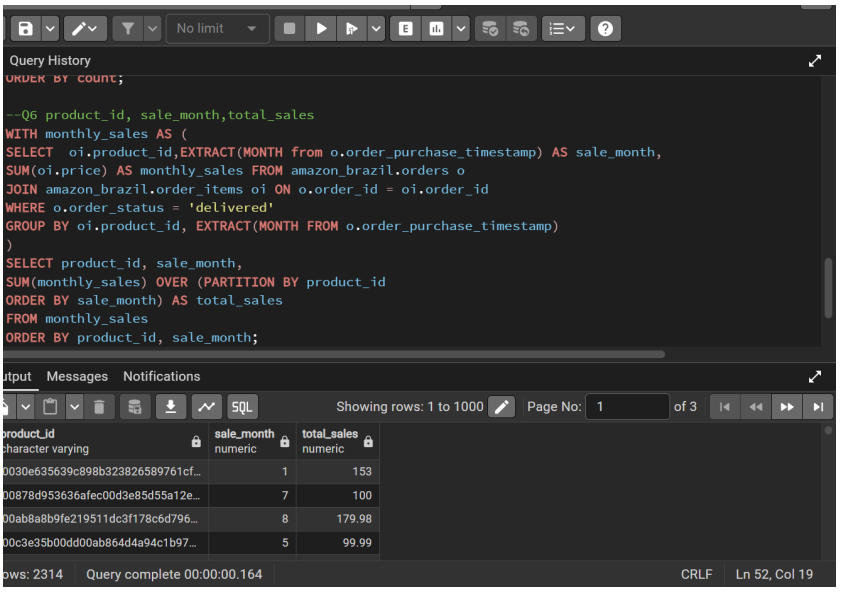
Strategy:-

- Personalized product recommendations

Question 6: Calculate monthly sales for each product starting from its first sale date.
Use a recursive CTE to compute cumulative sales (running total) month by month for each product.Display the cumulative sales trend per product.

Context:Amazon wants to analyze how sales grow over time for key products across their lifecycle. Tracking cumulative monthly sales helps identify product growth patterns, maturity stages, and long-term performance trends.

SQL Query (PostgreSQL) snippets:



Business Insights:

Different products peak in different months, indicating:

- Seasonality
- Demand cycles
- Promotion-driven sales

Some products sell better in certain months.Example:

- Electronics during **holiday seasons**
- Home goods during **moving seasons**

Some products show consistent growth over months.

These may be:

- Trending products
- Products benefiting from marketing campaigns

Recommendation

Identify the products with **highest cumulative sales** and:

- Promote them more
- Use them in advertising campaigns
- Bundle them with slower products

maximize revenue from proven performers

Products with very low sales across months might be:

- overpriced
- poorly marketed
- irrelevant to customer